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Rethinking Output Disclosure Control for Qualitative Data in Secure Data Facilities

International Secure Data Facility Professionals Network (ISDFPN)

17 April 2024

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Mitglied der Leibniz Gemeinschaft

Output Disclosure Control Challenges

- Few rules for de-identifying qual compared with quant (ONS has ODC done by people who are not methods experts)
- Range of methods and data formats greater than in quant (at least vis-à-vis problem disclosure)
- Big problem is context as disclosure risk – e.g., others who know the participant, events
- Growth of AI (Named Entity Recognition, Large Language Models) - risks AND solutions

Can we learn from curation methods?

- Key point: for much qual, there is no difference between „raw“ and „output“, except quantity
- Can view outputs as incoming data, **AS IF** the task were to prepare data for „Safeguarded“ access
- What is this task during curation?
  - Alter „raw“ sensitive data to reduce identification risk, thus available for less restricted access
- What is the task of output disclosure review?
  - Review small quantities of analysed data, to determine, then mitigate, disclosure risk
- So what? We ***do*** know how to do this…

Atypical, but instructive, example I

- 32 interviews; sexual assault; funder mandated sharing
- Litigated cases, accounts were already „public“
- 3 Qs: **Who** else? (perpetrator) **How**? (direct/observe) **What** other records? (FOIA, court records)
- Review not possible by victims, but agency advocates
- Standard techniques: de-ID, blur/aggregate, redact
- Plus: redaction + researcher summary

Campbell R, Javorka M, Engleton J, Fishwick K, Gregory K, Goodman-Williams R. Open-Science Guidance for Qualitative Research: An Empirically Validated Approach for De-Identifying Sensitive Narrative Data. Advances in Methods and Practices in Psychological Science. 2023;6(4). doi:10.1177/25152459231205832

Atypical, but instructive, example II

- Surprises: „kidnap and rape at gunpoint“ was NOT disclosive in this location, because so frequent *(deep knowledge can permit LESS redaction)*

- Despite heroic effort „some of the deepest, richest data were redacted“ (*still need Safe Facilities*)

- Did not seek waiver: „research participants explicitly said they wanted others to learn from their experiences…“ *(don‘t forget the power of consent)*

College and Beyond II

- Impact and outcomes of higher education-US Students
  - 19 institutions, 2000-21, (survey and open text responses)
- Virtual Data Enclave/ICPSR „*Guidelines for Protecting Confidentiality*“
  - Written by: CBII PI, research asst, and ICPSR's Privacy and Security Officer
  - Researchers can publish „qualitative data segments“
  - 8 sentence limit; in pilot 6 sentences were avg, +2 cushion!
  - Some data cannot be shared (indicated by quant data), names of campuses, course numbers, specifc majors
  - ICPSR Disclosure Review–usually done by ICPSR staff, but here, only PI and and ICPSR's Privacy and Security Officer

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Qual Disclosure Review Request Form (CBII)

[Note: The following data are fictional and do not actually appear in the CBII data. They are provided for illustrative purposes only]

A	B	C	D	E	F
id_person	Quotation of no more than 8 sentences, exactly as it appears in the data	Heightened risk of disclosure *(Y/N)*?*	If Column C=*Y*: Quotation of no more than 8 sentences as redacted by you. Please highlight the redaction.	Respondent characteristics you plan to indicate in your work that aren't mentioned in the quotation (limit 2, can't pose a heightened risk of deductive disclosure).	How does removing this quotation from the VDE help answer your research question or further the aim of your project?
100015853	The most important aspect of college for me were the connections I made in my fraternity at [CBII INSTITUTION]. A guy I knew in that fraternity also took ECON 143 with me, and we became pretty close. Later on he recruited me to work with him at [COMPANY], where I've become a leader in global management consulting.	Y (contains course number)	The most important aspect of college for me were the connections I made in my fraternity at [CBII INSTITUTION]. A guy I knew in that fraternity also took [ECONOMICS CLASS] with me, and we became pretty close. Later on he recruited me to work with him at [COMPANY], where I've become a leader in global management consulting.	Man	I plan to use this quotation in a manuscript to illustrate the theme of "networking", which was discussed more often by men than women in the open response data. My analysis of the data suggests the most impactful aspect of college often relates to career development for men, which is well illustrated by this quotation.

Scaling* a Solution

- Formalize training for qualified checkers-but does seem that people NOT from original project can do this (sometimes….)
- AI….because …..well…..AI……

Enabling qualitative research data sharing using a natural language processing pipeline for deidentification: moving beyond HIPAA Safe Harbor identifiers

Aditi Gupta[1], Albert Lai [1], Jessica Mozersky[2], Xiaoteng Ma[1], Heidi Walsh[2], and James M. DuBois [2]

JAMIA Open, 4(3), 2021, 1–10 doi: 10.1093/jamiaopen/ooab069 Research and Applications

- *But, the amount/volume of outputs to be checked is small (so far…), is automating ODC really top priority?

De-ID (formerly QuaDS Software)

- is a tool to help researchers de-identify qualitative data…names, ID nos, addresses, geo, age, org names, race/ethnicity, LGBTQ.
- exclusive licensing rights held by SCRC's The Institute for Mixed Methods Research. contact@immrglobal.org. (I have emailed)
- And others – Qualiservice, UKDS, QDR?

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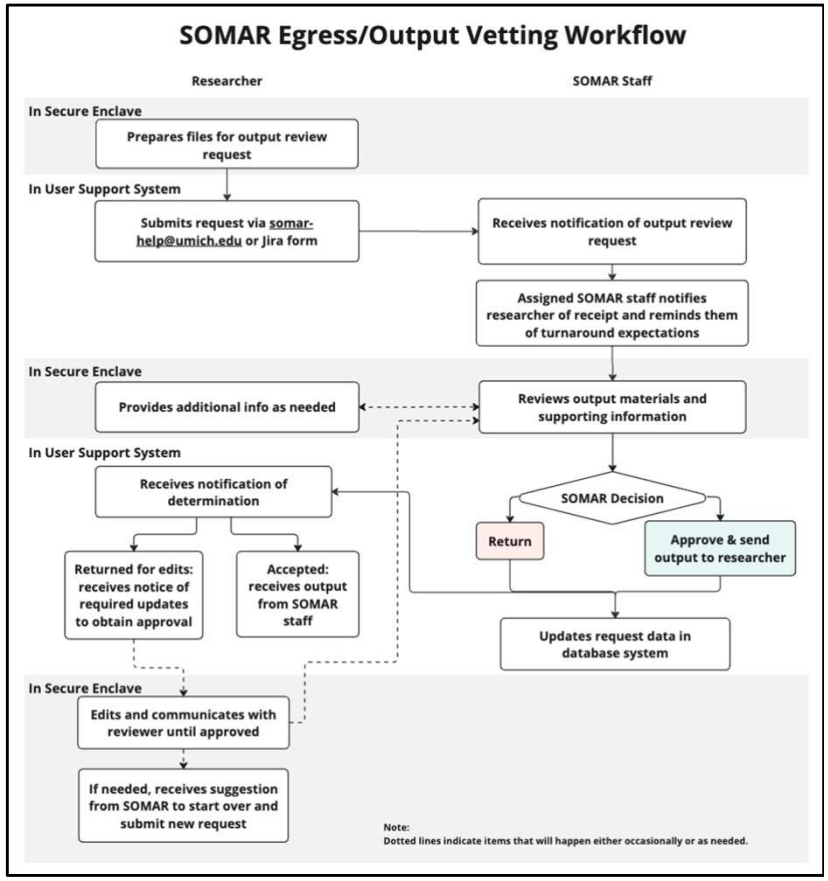
Next Steps I

- SOMAR–Social Media Archive-ICPSR
  - "Tweak" quant workflows for qual
  - No requests yet for output review with qual data, but they have a process drafted, based on ICPSR standards
- ICPSR-qual data in general
  - "Implementing advanced strategies to determine the risk of re-identification through indirect identifiers"
  - using expert determination of disclosure risk by one or more staff with training and expertise in disclosure risk review of qualitative output. There is also a Disclosure Review Board (made up of ICPSR staff and external researchers) should ICPSR staff want additional advice



- Categories/formats of qual data
  - Interviews, focus groups, et al.
- Demographic and contextual risks

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Next Steps II

- Contact qual projects that may do output reviews
  - Email sent to American Voices Project
  - (from S Karcher-tbc) allow use of approved direct quotations; use a heuristic version of k-anonymity to approve individual quotes; prohibit information that places respondents in a "demographic cell" that represents less than 10,000 people in the U.S. Census. The US Census itself takes a similar approach (AVP may use their rule on the Census paper).

- *National Child Development Survey* –
  - Call next week with Jane Elliot

- Qualiservice – German RDC (Kati Mozygemba)
  - QualiAnon for anonymizing qual data

Thanks to…

- Sebastian Karcher Qualitative Data Repository (QDR)
  - sensitive data, via Data Use Agreement, but no ODC
  - https://qdr.syr.edu/policies/sensitive-data
- Allyson Flaster *College and Beyond II*
- ICPSR
  - SOMAR Social Media Repository-Alison Sweet
  - Amy Pienta
  - Maggie Levenstein
- Qualitative Data Sharing (QDS) Project Series
  - https://www.icpsr.umich.edu/web/ICPSR/series/1780
- Rebecca Campbell (for reminding me why this work is worth the effort….)

“That which hath made me drunk hath made me bold.”
Lady Macbeth

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Mitglied der Leibniz Gemeinschaft

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Curation for access/ODC for publication

	Archive	ODC for PUblicaton
Char of data	Full data collection	extract
	"Raw"	Usually semi processed, but not necessarily (quote from interview is raw)
	Can be large	Usually Very small (meaning manual review feasible?)
Processing options	More detailed (e.g. full DOB, exact salary – depending on access route)	Some aggregation/grouping/top coding, etc. of variables
Who can access	Can restrict in TRE, non-commercial use; academic use, PhD & above, etc.	Fully public, searchable, Usually no restrictions, or just light-touch like registration
Who can assess risk	Data owner and archive staff have relevant expertise (but usu owners make decisions)	Key question – does this have to be data owner? Might vary for diff kinds of data…
	SDC checks (SDC Micro, etc.)	Ex. College and Beyond rules – once written, could another entity (archive?) enforce? Or at least handle most cases, and refer close calls to data owners?
Decision on access conditions	Varies depending on detail – download after reg, download with contract, within TRE	Through web browser or download
Level of service	Long term curation	Upload and leave ?? (ask Debs)